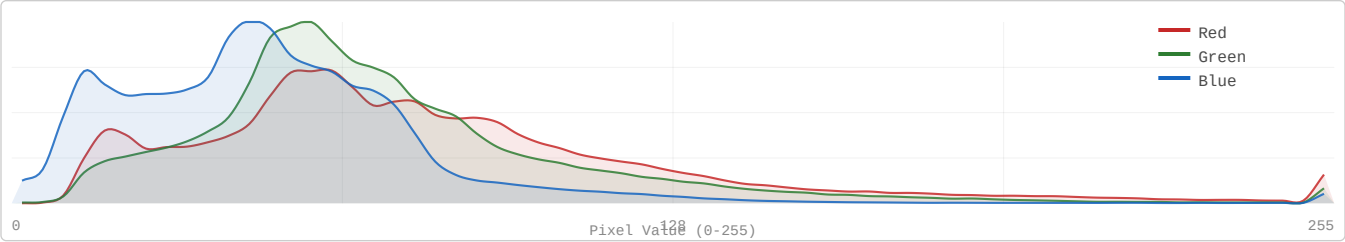
Decorrelation Gap: KODIM18

512x768 | 24-bit RGB | PNG (lossless)

Still Life with Fruit and Pottery | Weakly one-dimensional (PC1 = 90.63%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

	R	G	B
R	1.0000	0.9158	0.7526
G	0.9158	1.0000	0.8352
B	0.7526	0.8352	1.0000

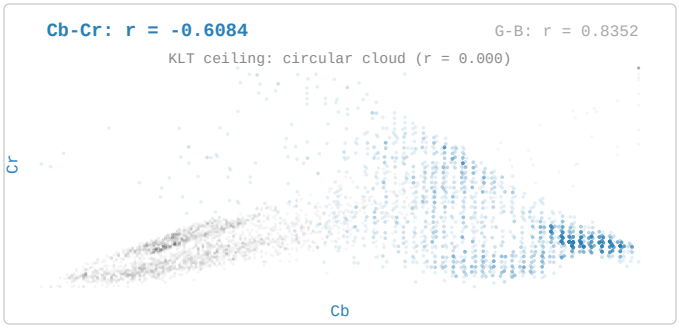
Cb-Cr: -0.608

(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	1.0000	-0.6075	0.4832
Cb	-0.6075	1.0000	-0.6084
Cr	0.4832	-0.6084	1.0000

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.915823	-0.607456	0.6075
R-B → Y-Cr	0.752554	0.483208	0.4832
G-B → Cb-Cr	0.835186	-0.608356	0.6084
Avg |r|	0.834521	0.566340	0.5663

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5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

32.1% Low

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

25.1%

0% Fully predictable — 100% Fully independent

6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	43.18 dB
G	47.87 dB
B	39.79 dB
Overall	42.48 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition

Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.7117	-0.5692	-0.4116	4188.57
PC2	0.5547	-0.0959	-0.8265	324.07
PC3	-0.4310	0.8166	-0.3840	108.98

8. PCA Derived Metrics

Condition Number

38.44

PC1/PC2: 12.92 | PC2/PC3: 2.97

Blue Channel Independence

25.1%

Variance not captured by PC1

PC1 Dominant Channel

R

Loading: R (0.71)

9. Redundancy Profile

Classification: Weakly one-dimensional | CN: 38.44 | PC1: 90.6%

RGB Correlation: Avg |r| = 0.834521 | R-G: 0.9158 | R-B: 0.7526 | G-B: 0.8352

YCbCr Residual: Avg |r| = 0.566340 | Y-Cb: -0.6075 | Y-Cr: 0.4832 | Cb-Cr: -0.6084

Efficiency: 32.1% of KLT ceiling | Gap: 0.5663 avg |r| remaining

Blue Channel: 25.1% independent | 4:2:0 PSNR: 42.48 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.